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Tom Y. Wu

Chapel Hill, NC, US   tomwu@unc.edu   tomwurdfz.github.io

Education

Ph.D **University of North Carolina at Chapel Hill**, Physics — Chapel Hill, US, Aug 2025 -
- Advisor: Prof. Carl Rodriguez
- Research Interest: Gravitational Waves, Compact Objects, Astrodynamics

Hons. B.Sc **University of Toronto**, Astrophysics and Computer Science — Toronto, Canada, Sep 2021 - May 2025
- Advisor: Prof. Maya Fishbach
- Degree: Astronomy and Physics Specialist, Computer Science Major
- Graduated with High Distinction

High School **High School Affiliated to Renmin University of China (RDFZ)** — Beijing, China, Sep 2018 - Jul 2021
- Top High School in China

First Author Publications

Are Long Gamma-Ray Bursts Progenitors to Merging Binary Black Holes? — **ApJ** 2024
T. Y. Wu, M. Fishbach
10.3847/1538-4357/ad98ed

Co-authored and Collaboration Publications

Implications of low neutron star merger rates for gamma-ray bursts, r-process production and Galactic double neutron stars — 2026
M.Fishbach, A.P.Ji, W-F.Fong, **T. Y. Wu**, J.C.Rastinejad, A.Vijaykumar, H-Y.Chen
10.48550/arXiv.2604.05059

GWTC-4.0: Population Properties of Merging Compact Binaries — 2025
The LIGO Scientific Collaboration et al.
10.48550/arXiv.2508.18083
Contribution: calculated binary neutron star merger rates.
News Coverage by CITA

Upper Limits on the Isotropic Gravitational-Wave Background from the first part of LIGO, Virgo, and KAGRA's fourth Observing Run — 2025
The LIGO Scientific Collaboration et al.
10.48550/arXiv.2508.20721
Contribution: computed binary neutron star merger rates under specified spin-distribution assumptions.

Contributed Talks

16th International LISA Symposium, MD, US — 2026
- Abstract accepted for a contributed talk

Research Experience

Graduate Student Researcher, University of North Carolina at Chapel Hill — Chapel Hill, US, Aug 2025 –
- Advisor: Prof. Carl Rodriguez
- Using Cluster Monte Carlo (CMC) and COSMIC to study the impact of intermediate-

mass black holes (IMBHs) in globular clusters.
- Investigating how IMBHs affect compact-object populations, binary black hole mergers, and extreme mass-ratio inspirals (EMRIs)relevant to LISA.

Undergraduate Researcher, University of Toronto — Toronto, Canada, Aug 2023 – Jun 2025

- Advisor: Prof. Maya Fishbach
- **Long gamma-ray bursts and binary black holes:** Estimated merger delay-time distributions, formation efficiencies, and spin-dependent effects.
- **Binary neutron star merger rates:** Analyzed merger-rate constraints under different astrophysical models and assumptions, using information from transients and Galactic binary neutron stars.
- Resulted in one first-author publication and one additional manuscript in preparation.

Undergraduate Researcher (Summer), University of Toronto — Toronto, Canada, May 2024 – Sep 2024

- Advisors: Prof. Ziqing Hong and Prof. Miriam Diamond
- Simulated detector-electronics behavior for the SuperCDMS experiment using computing clusters.
- Developed a pipeline to compare simulations with acquired data and improve detector-response modeling.
- Presented this work at the Canadian Astroparticle Physics Summer Student Talk (CASST) at SNOLAB.

Summer Research Exchange Student, National University of Singapore — Singapore, May 2023 – Jul 2023

- Advisor: Prof. Alexey Berdyugin
- Studied the use of machine-learning techniques to identify and map two-dimensional materials such as graphene.
- Worked on software and hardware tools to automate the graphene flake-search process.

Awards and Scholarships

Travel Grant - 16th International LISA Symposium — 2026

- $1295 USD from NASA

University of Toronto - Dean's List Scholar — 2022, 2023, 2024, 2025

Victoria College - The Regents In-course Scholarship — 2022, 2023, 2024

Undergraduate Summer Research Exchange Award — 2023

University of Toronto Admissions Scholarship — 2021

The International Mathematical Modeling Challenge — 2020

- Global Meritorious Award, Top 4 among 54 invited teams from 30 countries.
- Greater China Region Outstanding Award, Top 8 among 650 teams.